Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Need for statistics: purposes, and the collection and tabulation of data. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	9
Topic	Need for statistic
Subtopic	Need for statistics: purposes, and the collection and tabulation of data
Theme	—
Curriculum objectives	• explain the purpose of statistic • analyse and interpret statistical data presented in tables
Previous knowledge	• Students are familiar with counting and recording information.
Lesson sequence	• explain the purpose of statistic — the teacher introduces the concept of statistics as a method of collecting and organising information • analyse and interpret statistical data presented in tables — the teacher takes the students through collecting data, recording it with tally marks, and setting it out in a frequency table
Instructional materials	• classroom register, bottle tops, pencils or counters.
Evaluation	• (a) Define statistic • (b) State three methods of collecting data. • (c) Prepare a frequency table for the following data: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2.

Lesson objectives

1.	explain the purpose of statistic
2.	analyse and interpret statistical data presented in tables

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Need for statistics: purposes, and the collection and tabulation of data

Class: JSS1 **Subject:** Mathematics **Week:** 9

Section 1: What Statistics Is For

When you count how many of your friends have dogs, or how many days it rained last month, you are gathering information. That information is called data. Data are facts or information collected for a particular purpose.

Statistics is the science of collecting, organising, presenting, analysing and interpreting data. It is the whole process that turns a pile of scattered facts into something useful. Instead of just gathering numbers and leaving them in a messy list, statistics helps you work with them so you can actually understand what they mean.

Statistics matters because it helps us to:

- collect information — gather the facts we need;
- organise information — sort and arrange the facts so they are easier to read;
- compare information — put facts side by side to see how they differ;
- make decisions — choose what to do based on the facts;
- present information clearly — show the facts so others can understand them.

For example, a school records each learner's name in a register. That is collecting data. If you count how many learners are in each class and write the totals in a table, you are organising and presenting the data clearly. The table lets you compare classes and decide, say, which class has the most learners.

Let's see statistics at work with a small example.

Aminu asks five friends how many brothers each of them has. The answers are: 2, 1, 3, 1, 2.

- Collecting: Aminu asked the friends and wrote down their answers — the facts were gathered.
- Organising: He sorts them into a table, listing each number and how many friends gave it. So "1" appears twice, "2" appears twice, and "3" appears once.
- Presenting clearly: He now sees the organised table at a glance, so anyone reading it understands the answers without re-asking.
- Comparing and deciding: He can compare the answers and see that more friends have 1 or 2 brothers than have 3. If he needs to guess a typical answer, he has the facts to decide on.

Practice

1. In your own words, what is statistics?
2. What are data?
3. Name three ways statistics helps us, and give one example of your own for each.

Section 2: From Collected Data to a Frequency Table

Before you can make sense of a set of facts, you need to gather them first. In statistics, the facts we collect are called **data**, and data can be collected in four main ways:

1. **Observation** — watching and noting what happens.
2. **Counting** — counting objects one by one.
3. **Interview** — asking people questions and writing down their answers.
4. **Questionnaire** — giving people a written list of questions to fill in.

For example, if you want to know which vehicles pass your school gate in one morning, you might watch and count each one as it goes by. That is collecting data.

Once you have collected your data, you need to record it clearly as it comes in. A handy way to do this is with **tally marks**. A tally is a small stroke you make every time an item appears. You make four upright strokes, and the fifth stroke crosses the four to make a bundle of five, like this: ||||. This makes the final count easy to read.

A **frequency table** is a table that shows how many times each item occurs in a set of data. Each different item is listed once, and next to it you write its **frequency** — the number of times that item appears. Let's see how a set of data becomes a frequency table.

Example — collecting data with tally marks

One morning you note the vehicles that pass your gate in this order:

bus, car, car, car, lorry, bicycle, bicycle, car, car, lorry, bicycle, car

Go through the list from the start. Each time you see a vehicle, add one tally mark beside its name. When you reach five marks for one vehicle, bundle them as a crossed group of five, and start a new group.

Car appears at positions 2, 3, 4, 8, 9 and 12 — six times, so it gets five marks bundled plus one extra: |||| |
Bus appears once: |
Lorry appears at positions 5 and 10 — twice: ||
Bicycle appears at positions 6, 7 and 11 — three times: |||

Now count each group of tally marks and write the result in the **Total** column. This gives the tally table:

Vehicles	Tally	Total
car		
bus		
lorry		
bicycle		
Motor Bike		

The **Total** column is the frequency of each vehicle — how many times it appeared in your data. You can check your work by adding the totals: $6 + 1 + 2 + 3 = 12$, which matches the twelve vehicles you collected.

Example — a frequency table

When you already have the collected data and want to know how often each item occurs, you set it out in a frequency table. Suppose a class of students is asked their favourite colour, and their answers are counted. The frequency table records each colour once and shows how many students chose it:

Colour	Frequency
Red	18
Blue	20
Green	14
Yellow	16
White	15
Pink	30

The frequency of Blue is 20, which means 20 students chose blue as their favourite colour. The frequency of Pink is 30, the highest, so pink was the most popular. Reading a frequency table like this lets you analyse and interpret your data at a glance.

Practice

1. State three methods of collecting data.
2. Prepare a frequency table for the following data:
   1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2

Check your work

Attempt every question before you read this page.

What Statistics Is For

1. Statistics is the science of collecting, organising, presenting, analysing and interpreting data.

   Recall the full meaning: statistics is not just gathering numbers — it is the whole science of collecting, organising, presenting, analysing and interpreting data.
2. Data are facts or information collected for a particular purpose.
3. Any three of: collect information, organise information, compare information, make decisions, present information clearly. Examples should match the ones you chose (for example, "compare information" — checking which month had the most rainfall).

   Look back at the list of five uses of statistics. Choose any three and think of your own everyday example for each, such as counting books in your bag (collecting) or writing a table of scores (presenting clearly).

From Collected Data to a Frequency Table

1. Any three of: observation, counting, interview, questionnaire.
2. | Number | Frequency |
   |---|---|
   | 1 | 3 |
   | 2 | 3 |
   | 3 | 4 |
   | 4 | 3 |

   Count how many times each number appears. 1 appears at positions 1, 2 and 12 — three times. 2 appears at positions 7, 8 and 13 — three times. 3 appears at positions 3, 6, 9, 10 — four times. 4 appears at positions 4, 5, 11 — three times. Check by adding: $3 + 3 + 4 + 3 = 13$, which matches the thirteen pieces of data.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-4d425202	**Date**	
Subtopic	Need for statistics: purposes, and the collection and tabulation of data	**Rater**	

The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____